

BSCE BIOCHM REVIEW

1. A 42-year-old woman comes to her primary care physician because of tingling, numbness, and weakness in her legs and feet. The symptoms started several weeks ago and affect both the left and right sides. She has a history of hypertension controlled with a thiazide diuretic, stable Crohn disease (after having two ileal resections more than 2 years ago), and depression, which is being treated with a selective serotonin reuptake inhibitor. She is concerned that her current symptoms might be the result of a stroke related to her high blood pressure. Which of the following is the most likely underlying cause of this patient's neurologic symptoms?

- ☐ (A) Cobalamin Deficiency
- ☐ (B) Diabetic Neuropathy
- ☐ (C) Hypercalcemia
- ☐ (D) Hypokalemia
- ☐ (E) Lacunar Infarct

2. A 34-year-old homeless man comes to the clinic after being punched in the jaw during a bar fight. On review of systems, he reports that his gums have been bleeding without an obvious cause during the past several months. He takes no medications. On examination, he is emaciated, and several bruises in different stages of healing are noted over his legs. He also has a history of anemia. Results of laboratory studies are as follows: Hemoglobin: 12.8 g/dL Prothrombin time: 14 sec Partial thromboplastin time: 29 sec Bleeding time: 2 min International normalized ratio: 1.1 Alkaline phosphatase: 54 IU/L Alanine aminotransferase: 20 IU/L Aspartate aminotransferase: 22 IU/L

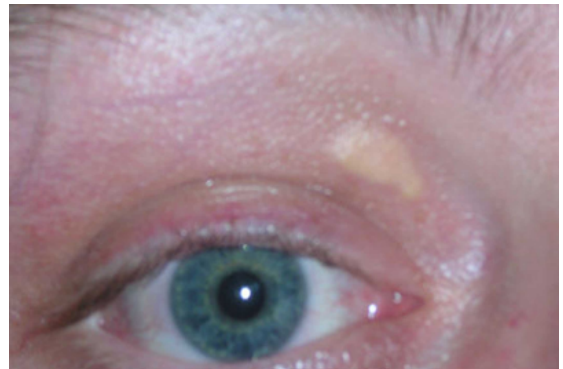
- ☐ (A) Decreased vWF
- ☐ (B) Decreased carboxylation of coagulation factors
- ☐ (C) Defect in hydroxylation of collagen residues
- ☐ (D) Dysfunctional platelets

- 3.** A 3-week-old boy is brought by his mother to the physician because he has been vomiting since birth. The mother is concerned because she thinks her son is “shrinking,” and she describes how he seems to be unable to keep his bottled milk feedings down. The infant was born at home and did not undergo newborn screening. His temperature is 37°C (98.6°F), pulse is 150, respirations are 35/min, and blood pressure is 70/50 mm Hg. He is at the 50th percentile for height and the 30th percentile for weight. On physical examination, the patient is found to have yellow discoloration of the skin, clouding of both eyes, a distended abdomen, and palpable liver margins.

- ☐ (A) Aldolase B
- ☐ (B) Fructokinase
- ☐ (C) Aldolase Reductase
- ☐ (D) Galactose 1 phosphate uridylyltransferase
- ☐ (E) Galactokinase

- 4.** A 2-year-old girl with findings similar to those shown in the image is brought to the clinic. On physical examination, the patient also has opaque rings on the lower edge of her irises. Laboratory test results include the following: triglycerides, 80 mg/dL; HDL, 70 mg/dL; and LDL, 750 mg/dL. A primary defect or deficiency in which of the following is most likely in this patient?

- ☐ (A) Apolipoprotein C II
- ☐ (B) Apolipoprotein E
- ☐ (C) LDL Receptors
- ☐ (D) Lipoprotein Lipase
- ☐ (E) VLDL Production



5. A 31-year-old white man presents to his primary care physician because of progressive pain and stiffening of his shoulder. He says that over the past 2 years he has noticed increasing difficulty throwing a baseball while playing in a local recreational league. His medical history is notable for chronic lower back and right shoulder pain treated with nonsteroidal anti-inflammatory drugs (NSAIDs). His back pain does not radiate to his legs, and he has no urinary or bowel incontinence. He has noticed progressive darkening of his ears and eyes over the past few years. His pulse is 72/min, respiratory rate is 18/min, and blood pressure is 138/84 mm Hg. He is afebrile. Findings on physical examination are remarkable for brownish gray pigmentation of his sclerae and pinnae. There is no joint warmth or erythema, but he does have stiffness and limited range of motion of his shoulders and lumbar spine. A systolic murmur is heard in the second right intercostal space on auscultation. Which of the following should be limited as part of the management of this patient's condition?

- ☐ (A) Tyrosine
- ☐ (B) Homocysteine
- ☐ (C) Cysteine
- ☐ (D) Leucine
- ☐ (E) Vitamin D

6. A 2-year-old boy is brought to the physician by his parents due to concerns about progressive hearing loss. Developmental progress has plateaued since his last visit approximately 9 months ago. The child exhibits a prominent forehead and enlarged tongue. Hepatosplenomegaly and an abdominal hernia are also noted on physical examination. Laboratory analysis shows a deficiency in iduronate sulfatase. Which of the following diseases is this patient likely suffering from?

- ☐ (A) Gaucher Disease
- ☐ (B) Hunter Disease
- ☐ (C) Krabbe Disease
- ☐ (D) Tay-Sach's Disease
- ☐ (E) I Cell Disease

7. A 6-year-old boy is brought to the emergency department by his parents after complaining of left knee pain and refusing to bear weight on the affected limb. The patient also complains of mouth pain, and an oral examination reveals inflamed and bleeding gums. The child is unkempt and has multiple abrasions on his elbows and knees that appear to be healing poorly. He has petechiae on his chest and back. Laboratory studies show a hemoglobin concentration of 10.2 g/dL. The patient's mother reports that they have been homeless and living in a shelter for the past year and that their food intake has been poor for the past 2 to 3 months. Which of the following is most likely responsible for this patient's symptoms?

- ☐ (A) Factor 8 Deficiency
- ☐ (B) Deficiency in carboxylated coagulation factors
- ☐ (C) Deficiency in collagen synthesis
- ☐ (D) Impaired free radical scavenging
- ☐ (E) Impaired immune response

8. A 17-year-old boy was recently prescribed sulfamethoxazole and trimethoprim (SMX/TMP) for a urinary tract infection. The patient calls the physician's office 3 days later and reports fatigue, darkening urine, and a yellow tinge to his skin and eyes. Laboratory tests are ordered and show the following: leukocyte count, 9100/mm³; hemoglobin, 10 g/dL; hematocrit, 34%; platelet count, 310,000/mm³. Which of the following compounds builds up because of this patient's condition?

- ☐ (A) Hemoglobin
- ☐ (B) Oxidized glutathione
- ☐ (C) Oxidized NAD⁺
- ☐ (D) Reduced glutathione
- ☐ (E) Reduced NADPH

9. A mother brings her 3-year-old son to the pediatrician because she is concerned about his increasing weakness and clumsiness. Although he met all of his early motor milestones as a baby, the mother recalls that he was not able to walk on his own until 18 months of age. He is now having increasing difficulty keeping up physically with his peers during playtime. On examination, the child's cognitive development is appropriate for age. Examination of the patient's lower extremities reveals normal patellar and Achilles reflexes, normal sensation, and 3/5 muscle strength. His calf muscles are enlarged (see image for example). Biopsy of the muscle reveals wide variation in the diameter of individual muscle fibers, with degenerative necrosis of some and interspersed regenerating fibers. Although the mother has not experienced the symptoms seen in her son, she states that she had a "sickly" brother who died at a young age after being confined to a wheelchair for some time. Her husband does not have any history of muscle weakness in his family. The mother is now 30 weeks pregnant with a baby girl and is concerned that this child may have the same health issues as her brother.



- (A) 25%
- (B) 50%
- (C) 100%
- (D) 0%
- (E) 75%

10. A 48-hour-old baby boy was born via spontaneous vaginal delivery to a G1P1 mother. The infant is irritable, is feeding poorly, vomiting, and shows lethargy and dystonia. His heart rate is 138/min, blood pressure is 84/53 mm Hg, temperature is 37°C (98.6°F), respiratory rate is 38/min, and oxygen saturation is 100% on room air. The patient's mother received the recommended prenatal care and also elected not to breastfeed after delivery. Laboratory studies were ordered and revealed an increase in plasma leucine and alloisoleucine. Which of the following is most likely to be deficient in this patient?

- (A) Alpha ketoacid dehydrogenase
- (B) Cystathione Beta-synthase
- (C) Homogentistic acid oxidase
- (D) Phenylalanine Hydroxylase
- (E) Positively charged amino acid transporter

11. A 31-year-old woman is trying to get pregnant. She has a niece who suffers from a genetic disease characterized by recurrent respiratory infections and pancreatic failure. She would like to assess her chances of having a child with this disease. Which of the following laboratory techniques is most appropriate to determine if this woman and/or her husband is a carrier of the mutant gene?

- ☐ (A) ELISA
- ☐ (B) Gel Electrophoresis
- ☐ (C) Karyotype Analysis
- ☐ (D) Northern Blot
- ☐ (E) PCR

12. A 6-year-old girl exhibits visual abnormalities during a screening test at school and is referred to an ophthalmologist to be fitted for glasses. On examination, the ophthalmologist notices that her right lens is dislocated downward, which, combined with her above average height, awkward gait, and scoliosis, makes the physician suspect an underlying condition. Which of the following substances would most likely be elevated in this patient's urine?

- ☐ (A) Leucine
- ☐ (B) Homocysteine
- ☐ (C) Phenylalanine
- ☐ (D) Tyrosine
- ☐ (E) Tryptophan

13. A male infant is born to a 22-year-old woman, G1P1, at 35 weeks' gestation. Vital signs are within normal limits. The infant is found to have an abnormally large cranium, a small jaw,, and low-set ears (see image). The infant subsequently dies of neurologic complications at 9 months of age. Which of the following is the most likely karyotype of this infant?

- ☐ (A) 47, XY Trisomy 21
- ☐ (B) 47, XY Trisomy 13
- ☐ (C) 47, XY Trisomy 18
- ☐ (D) 46, XY microdeletion at 7p
- ☐ (E) 46, XY microdeletion at 22q.11

- 14.** An 18-month-old girl with a rare hereditary disease is brought to a genetics clinic by her parents. She is small for her age with club feet, coarse facial features, and enlarged gums. She is unable to roll or sit unsupported. Her parents note that she was 17 months old when she said her first word and report that her voice is constantly hoarse. The geneticist explains that this child's disease is caused by the failure of a process that occurs in the cellular structure indicated in the image below. The organelle pictured is essential for which of the following cellular processes?



- (A) Beta oxidation of long chain fatty acids
 - (B) Degradation of cellular materials
 - (C) Lipid and membrane biosynthesis
 - (D) Protein Modification
 - (E) Protein Synthesis
- 15.** A 58-year-old man with a history of alcohol abuse comes to the hospital with a protuberant abdomen, slurring of speech, and confusion. Physical examination reveals multiple spider angiomas on his chest, and abdominal examination elicits a fluid wave. When his hands are extended outward, a flapping tremor appears. He is promptly administered medications to reduce the level of the toxic metabolite responsible for his symptoms. Normally, where do the initial steps for detoxification of this metabolite take place?
- (A) Hepatocyte cytoplasm
 - (B) Hepatocyte mitochondria
 - (C) Hepatocyte sER
 - (D) Kupffer cell cytoplasm
 - (E) Portal circulation

- 16.** A 7-year-old boy is brought to the pediatrician because of a history of chronic cough, greasy floating diarrhea, and failure to gain weight. The patient's body mass index is calculated at 17 kg/m², and he is in the 40th percentile for height. On auscultation, bronchial breath sounds are heard, and frequent sputum production is observed during the office visit. *Pseudomonas aeruginosa* is grown from the patient's sputum culture. The physician subsequently orders sweat testing, and the chloride content is found to be markedly elevated. Genetic testing reveals the most common mutation seen in this disease. Which of the following is the most likely underlying cause of this patient's condition?
- ☐ (A) Impaired ion conduction
 - ☐ (B) Impaired protein production
 - ☐ (C) Impaired protein trafficking
 - ☐ (D) Increased turnover of ion channels in the plasma membrane
 - ☐ (E) Reduced ion channel response to ATP stimulation
- 17.** An 8-year-old boy is brought to the clinic with a complaint of a runny nose and difficulty breathing. His mother says the boy has had many respiratory tract infections throughout his childhood, often with a productive cough. The child's symptoms are tolerable in the morning but progressively worsen as the day continues. Chest auscultation reveals crackles and wheezing, but the physician is unable to auscultate normal S1 and S2 heart sounds. Radiographic examination reveals large, dilated airways and a right-sided cardiac shadow.
- ☐ (A) Eisenmenger Syndrome
 - ☐ (B) Impaired fertility
 - ☐ (C) Panacinar emphysema
 - ☐ (D) Pancreatitis
 - ☐ (E) Pulsus paradoxus
- 18.** A 43-year-old man comes to the clinic complaining of a "burning" feeling on both forearms. He has noticed an accompanying rash, which he initially thought was a sunburn. He says that it is not itchy, just painful. He also mentions weakness, loss of appetite, loose stools, and increased anxiety. For the past 6 months, he has been homeless and eating whatever he can find. On physical examination, the affected skin appears thickened and hyperpigmented. A similar lesion is seen on the neck. The patient's tongue appears enlarged and erythematous. Findings from a neurologic examination are normal. The patient appears agitated, and at times, he struggles to continue the interview. Which of the following is most likely deficient in this patient?
- ☐ (A) Biotin
 - ☐ (B) Pantothenic Acid
 - ☐ (C) Thiamine
 - ☐ (D) Tryptophan
 - ☐ (E) Zinc

19. An 8-month-old boy is brought to the pediatrician by his parents because of general weakness. His parents state that he has progressively lost the ability to hold his toys and can no longer support his weight when crawling. On examination, the patient is tachypneic and breathing with considerable effort. The liver is palpable five finger widths below the right costal margin. The patient has a difficult time sitting upright, and he cannot squeeze the physician's fingers with any appreciable strength. A chest X-ray is performed and reveals an enlarged cardiac silhouette. Despite a number of interventions, the child's symptoms continue to worsen until his death 2 weeks later. This patient's disease is most likely caused by intracellular accumulation of which of the following substances?

- ☐ (A) Glucose
- ☐ (B) Glycogen
- ☐ (C) Oxaloacetate
- ☐ (D) Pyruvate
- ☐ (E) Urea

20. A 2-year-old girl is brought to the clinic because she has recently developed muscle weakness and is having difficulty walking. She was meeting her early developmental milestones as expected at her 18-month well-child checkup. Her physical exam shows signs of muscle wasting. A peripheral nerve biopsy reveals histologic evidence of demyelination and macrophages with prominent reddish-pink vacuoles in the cytoplasm after staining with toluidine blue. This patient has a deficiency of which of the following enzymes?

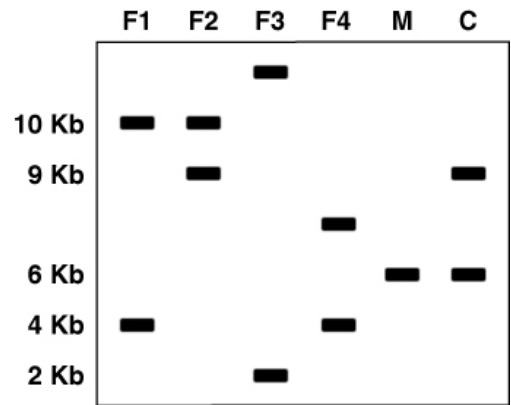
- ☐ (A) alpha-galactosidase A
- ☐ (B) beta-hexosaminidase A
- ☐ (C) arylsulfatase A
- ☐ (D) galactocerebrosidase
- ☐ (E) glucocerebrosidase
- ☐ (F) iduronate sulfatase

21. A 25-year-old mother with limited access to prenatal care gives birth to a child at 32 weeks' gestation with several birth defects. The baby is admitted to the neonatal intensive care unit for resuscitation and examination. Physical examination reveals a cleft lip, extra digits, and extremely small eyes. What is the most likely cause of these findings?

- ☐ (A) Deletion of 11p13
- ☐ (B) Deletion of 5p
- ☐ (C) Trisomy 13
- ☐ (D) Trisomy 18
- ☐ (E) Trisomy 21

- 22.** A 20-year-old mother, unsure of the paternity of her newborn son, requested a genetic test to make this determination. Samples from the four men suspected to be the father (F1, F2, F3, F4), the mother (M), and the infant (C) were collected, amplified, and treated with a restriction enzyme, EcoRI. The resulting fragments were separated with gel electrophoresis, and specific sequences of the DNA were identified later via hybridization. Which of the following techniques is best described by the analysis above?

- (A) Flow cytometry
- (B) Microarray
- (C) Northern Blot
- (D) Southern Blot
- (E) Western Blot



- 23.** A 15-year-old boy with intellectual disability is brought for his annual check-up examination. Family history shows that the child's maternal uncle has the same condition, but the child's case is more severe than his uncle's. Vital signs are within normal limits; physical examination shows enlarged testes. A photograph of the patient is shown. This is an example of which of the following specific genetic phenomena?

- (A) Anticipation
- (B) Heteroplasmy
- (C) Imprinting
- (D) Incomplete Penetrance
- (E) Locus Heterogeneity



24. An 8-day-old boy is brought to the pediatrician's office for a newborn checkup. His mother states that his gums have been bleeding after he breastfeeds and that she has seen bloody streaks in his diapers. He is otherwise healthy and has been eating, stooling, and voiding well. He had an uncomplicated vaginal birth at 40 weeks to a primigravid woman at home. He was breastfed exclusively, but recently has been formula-supplemented due to poor milk production. Family history is significant for neonatal jaundice in a cousin. He does not take any medications. On examination, his temperature is 37° C (98.6° F), blood pressure is 80/40 mm Hg, pulse is 160/min, and respiratory rate is 44/min. There is some mucosal bleeding on his gums and a small blood stain on the inside of his diaper where it covers his umbilicus. Laboratory results are significant for: Bleeding Time: Normal PT: 40 sec (High) INR: 2.1 PTT: 55 sec (High) ALT: 10 IU/L AST: 8 IU/L Which of the following is the most likely cause of this infant's presentation?

- ☐ (A) Congenital Factor VIII Deficiency
- ☐ (B) Consumption of coagulation factors
- ☐ (C) Immature gut microbiome
- ☐ (D) Impaired Von Willebrand factor function
- ☐ (E) Obstruction of pancreatic duct

25. A 7-year-old girl is brought to the dermatologist by her parents who are very concerned about her skin. They report that she has a new enlarging mole over her back. Her parents report that she has had severe, persistent sunburns, even with minimal sun exposure, since infancy. The girl has been otherwise healthy. On physical examination, her face, legs, and arms are covered with freckles and there is a 1-cm × 2-cm asymmetric nevus over her right scapula. Which of the following is the most likely cause of this patient's skin findings?

- ☐ (A) A defect in base excision repair
- ☐ (B) A defect in homologous recombination
- ☐ (C) A defect in mismatch repair
- ☐ (D) A defect in nonhomologous end joining
- ☐ (E) A defect in nucleotide excision repair

26. A 75-year-old man dies overnight in the intensive care unit after being there for the past 6 days. Since the cause of his illness and subsequent death is still unclear to the care team, the patient's family consents to the physician's request for an autopsy. On initiation of the autopsy the following morning, the muscles of the body are hard and very stiff, and it is difficult to move and manipulate his limbs. The inability of the muscle fibers to relax is due to specific biochemical changes within the muscle. Which of the following mechanisms is impaired, preventing normal movement?

- ☐ (A) Binding of actin to myosin
- ☐ (B) Binding of troponin to tropomyosin
- ☐ (C) Power stroke of the myosin head
- ☐ (D) Release of actin from myosin
- ☐ (E) Uptake of calcium into the sarcoplasmic reticulum

- 27.** A pediatrician is called to the hospital nursery to evaluate a 3-day-old boy for a suspected seizure. The boy was reported to have had a tonic-clonic seizure lasting for 40 seconds. Review of records shows the baby was born via cesarean section. The mother received full prenatal care. The baby is noted to have been born with a cleft palate. Additionally, he failed his newborn hearing test. Since birth, he has been having some "muscle jerking." Physical examination today shows a unilateral incomplete cleft palate. A 2/6 systolic murmur is detected and is best heard over pulmonic area. During the examination, the baby appears jittery. This child's deficits have most likely arisen from abnormal migration of cells from which of the following structures?
- ☐ (A) 1st and 2nd pharyngeal arches
 - ☐ (B) 1st and 2nd pharyngeal clefts
 - ☐ (C) 4th and 5th pharyngeal pouches
 - ☐ (D) 2nd and 3rd pharyngeal clefts
 - ☐ (E) 3rd and 4th pharyngeal pouches
- 28.** A 28-year-old African-American man presents with jaundice and dark urine 2 days after starting primaquine for his upcoming trip to sub-Saharan Africa. He denies any comorbidity and regular medication use. He recalls an older brother having a similar episode prior to his trip to the same region, but not his younger sister. On examination, his vital signs are stable and he is afebrile. There is mild conjunctival pallor and scleral icterus. There is no palpable lymphadenopathy or organomegaly. Laboratory tests show: WBC count: 8400/mm³ Hemoglobin: 12 g/dL Hematocrit: 36% Platelet count: 289,000/mm³ Production of which of the following molecules is decreased in this condition?
- ☐ (A) Free radicals
 - ☐ (B) Galactose 1-phosphate
 - ☐ (C) Pyruvate
 - ☐ (D) The reduced form of NADP⁺
 - ☐ (E) Tyrosine
- 29.** A 23-year-old man comes to the emergency department; he has extreme abdominal pain and rebound tenderness. The patient appears uncomfortable and is unable to sit still. On examination, his skin is thin and appears nearly translucent with prominent veins. Varicosities and bruises are scattered over his lower limbs. A stat CT scan of the abdomen reveals free fluid and air in the abdomen. The patient is rushed to the operating room for an emergent exploratory laparotomy and is found to have an intestinal rupture. Which of the following is most likely defective in this patient's disease process?
- ☐ (A) Fibrillin
 - ☐ (B) Type I collagen
 - ☐ (C) Type II collagen
 - ☐ (D) Type III collagen
 - ☐ (E) Type IV collagen

- 30.** A 2-day-old boy is evaluated in the emergency department after being found unresponsive. The mother reports her son has been lethargic and has been feeding poorly since birth. Today he has been more irritable than normal, and has had several episodes of vomiting. The infant's arterial pH is 6.8, partial pressure of carbon dioxide in blood is 25 mm Hg, and bicarbonate level is 16 mEq/L. A metabolic work-up is initiated. Plasma organic acid testing reveals elevated levels of leucine. A mutation in which of the following enzymes is most likely responsible for the infant's symptoms?
- ☐ (A) Alanine Aminotransferase
 - ☐ (B) Branched chain alpha-keto acid dehydrogenase
 - ☐ (C) Carbamoyl phosphate synthetase I
 - ☐ (D) Glutamate dehydrogenase
 - ☐ (E) Ornithine transcarbamylase
- 31.** A 42-year-old pregnant woman (gravida 2, para 3) woman at 40 weeks' gestation is brought to the hospital where she delivers her child vaginally without complication. During the infant's initial physical examination, the obstetrician notes upslanting palpebral fissures, epicanthal folds, and a depressed nasal bridge. The mother states that during the pregnancy she took prenatal vitamins as recommended and was careful to avoid substances that may have caused any harm to the developing fetus. Which of the following is the most likely cause of this infant's presentation?
- ☐ (A) Allelic heterogeneity
 - ☐ (B) Chromosomal deletion
 - ☐ (C) Germline mosaicism
 - ☐ (D) Maternal uniparental disomy
 - ☐ (E) Nondisjunction in meiosis I
- 32.** An Ashkenazi Jewish couple expecting their first child came to the physician. Both parents are healthy and have no medical conditions. The mother's family history is remarkable for a brother with cystic fibrosis; her parents are unaffected. The father's brother also has cystic fibrosis. Upon his brother's diagnosis, the father was tested and found to be a carrier of the Phe508 gene mutation. The mother has not been tested for genetic mutations. Which of the following represents the likelihood that this couple's child will be affected by cystic fibrosis?
- ☐ (A) 100%
 - ☐ (B) 17%
 - ☐ (C) 25%
 - ☐ (D) 3%
 - ☐ (E) 50%

- 33.** During an investigative research on epidermal basal cells, researchers identify non-coding, repetitive regions of DNA at the ends of chromosomes. Which of the following best describes the function of these genetic elements?
- ☐ (A) Alter transcription of linked genes
 - ☐ (B) Initiate transcription of single gene
 - ☐ (C) Maintain chromosome integrity
 - ☐ (D) Removal of a gene sequence in RNA processing
 - ☐ (E) Self excision and transfer of DNA segments
- 34.** A 30-year-old man is brought to the family physician by his wife, who has noticed that his behavior has become erratic. She reports that over the past 2 years, he has not “been himself” and that he has developed a slowly worsening depression. She also mentions that starting around 2 years ago, his speech became uncoordinated and his gait became abnormal. The patient has a hard time finding his balance. On examination, the physician notices asterixis and that the liver and spleen are palpable. The patient has dark annular deposits around both corneas, yellow skin, and a resting tremor in his right arm. His lab results show: Alanine aminotransferase: 234 U/L Aspartate aminotransferase: 346 U/L Alkaline phosphatase: 75 U/L Total bilirubin: 4.8 mg/dL Direct bilirubin: 3.9 mg/dL This patient’s disorder is caused by a disruption in which of the following processes?
- ☐ (A) Amino acid production
 - ☐ (B) Copper metabolism
 - ☐ (C) Iron excretion
 - ☐ (D) Lipid metabolism
 - ☐ (E) Mucopolysaccharide dysfunction

35.



An 8-year-old boy is brought to his physician because of severe pain in his left foot. He was running on the playground when he tripped and fell. He was unable to bear weight on his left foot immediately after the fall. His mother adds that her son also had a broken arm within the past 2 years. Results of a complete blood cell count and the erythrocyte sedimentation rate are normal. A picture of the child's eyes is shown. Which of the following is the most likely cause of this patient's disease?

- (A) Abnormal type I collagen synthesis
- (B) Abnormal type III collagen synthesis
- (C) Abnormal type IV collagen synthesis
- (D) Defect in fibrillin synthesis
- (E) Mutation in the tumor suppressor gene Rb

36. A mother brings her 5-year-old son to the pediatrician because he is not able to keep up with his friends when they are playing and running around. The boy's mother denies that he seems out of breath when he runs but reports that he has trouble moving from a seated position on the floor to a standing position. He has to push himself off of the floor or pull himself up in order to stand. Physical examination reveals a cheerful boy with normal height and weight, normal reflexes, and large calves. His plantar flexion strength is a 4/5 bilaterally. Which of the following is the most likely cause of this patient's condition?

- (A) A missense mutation
- (B) A nonstop mutation
- (C) A splice site mutation
- (D) Frameshift mutation
- (E) Nonsense mutation

- 37.** A 27-year-old man is brought to the emergency department with crushing substernal chest pain radiating to his back that occurred suddenly while lifting heavy weights at the gym. He denies any history of similar pain. His blood pressure is 85/25 mm Hg, pulse is 90/min, respiratory rate is 22/min, oxygen saturation is 92%, and he is afebrile. Physical examination reveals a tall, slender man with long arms and fingers, as well as new-onset, grade IV/VI diastolic decrescendo murmur heard best at the second intercostal space right sternal border. Which of the following is the most likely underlying cause of this patient's presenting symptoms?
- ☐ (A) Atherosclerosis
 - ☐ (B) FBN1 gene mutation
 - ☐ (C) Multiple endocrine neoplasia type 2B
 - ☐ (D) Obliterative endarteritis of the vasa vasorum
 - ☐ (E) Type III collagen defect
- 38.** A 64-year-old homeless man comes for evaluation because of a painful, erythematous laceration on his right calf. He reports having initially injured the area on a shard of glass approximately 3 weeks ago. He says that while the wound stopped bleeding initially, he reports that it never "scabbed over." On physical examination, there are multiple bruises on his body, petechiae, pallor, and swollen gums. At the cellular level, what is the dysfunction underlying this man's condition?
- ☐ (A) Decreased activity of thyroperoxidase
 - ☐ (B) Decreased hydroxylation of proline and lysine residues
 - ☐ (C) Decreased intestinal absorption of calcium
 - ☐ (D) Decreased synthesis of methionine
 - ☐ (E) Decreased transketolase activity
- 39.** A 3-year-old boy is brought to the pediatrician's office by his mother. She reports that the child has been healthy since birth but has recently begun to have difficulty standing. In the office, the physician notices that the child has enlarged calves and when he attempts to stand up and he uses his hands and arms to push himself to a standing position. After additional testing, he is diagnosed with a genetic condition. Which of the following disorders shares the same inheritance pattern as this patient's genetic disease?
- ☐ (A) Cystic fibrosis
 - ☐ (B) Hemochromatosis
 - ☐ (C) Hemophilia A
 - ☐ (D) Marfan Syndrome
 - ☐ (E) Neurofibromatosis Type 1

40. A pharmaceutical company is investigating the effect of an experimental drug on the murine cardiac hypertrophic response. This drug binds to a myocyte transmembrane protein and activates a G α_q subunit. This interaction increases cardiac inotropy and decreases the action potential refractory period. Increased levels of which of the following intracellular constituents are most likely responsible for this observed response?

- ☐ (A) cAMP
- ☐ (B) cGMP
- ☐ (C) IP₃
- ☐ (D) NO
- ☐ (E) PKA
- ☐ (F) Na⁺